

One CSI Report Cannot Serve Every Consumer: The Consumer-Relative False-Clear Rate in 5G/6G Link Adaptation

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Abstract—A CSI report carries a promise: the channel will support the rate selected from it. Whether the promise holds depends on who relies on it. We measure how often it fails, on a 5G NR link with real LDPC decoding, and show that the failure rate depends on the reader in two ways. First, the reliability target. An MCS choice that meets an eMBB error target of 10^{-1} misses a URLLC target of 10^{-3} on the same channel by a factor of 112, and it does so at every SNR from cell edge to cell centre. A policy that knows the stricter target meets it. Second, time. As a report ages the failure rate grows, and the longest safe report period is about 0.177 of the Clarke coherence time ($R^2 = 0.915$), until at high mobility no period the frame permits is short enough. The corrections are standard, outer-loop link adaptation and faster reporting. What we add is the measured failure rate that says how much each correction is worth and when it stops being optional. A third experiment shows the failure rate can invert: two margin policies with equal budgets, and each of two user fleets prefers a different one, so no fleet-level number serves both. Every number comes from a sealed, pre-registered record.

Index Terms—CSI feedback, link adaptation, URLLC, outer-loop link adaptation, reliability, 5G NR.

I. INTRODUCTION

Adaptive modulation and coding turns a CSI report into a rate decision. The transmitter selects the highest MCS whose required SINR the report clears. The report therefore carries an implicit claim: the link will decode at the chosen rate. We call the optimistic failure of that claim a false-clear. The CSI cleared an MCS the channel cannot support, and the block is lost.

Whether a report false-clears is not decided by the channel alone. It is decided by the report together with whoever acts on it. Two schedulers can read the same CSI and disagree, because they hold traffic to different requirements. This paper measures that disagreement on a real 5G NR link.

The disagreement shows up along two axes. The first is the reliability target (§III). An MCS that serves eMBB adequately misses a URLLC target on the same channel by more than a hundred times, and the gap persists at every SNR we tested. The second is time (§IV). As the report ages the false-clear rate grows, and the longest safe report period is a fixed fraction of the channel coherence time, until mobility is high enough that no period is short enough.

Neither observation needs a new estimator or scheduler. The corrections are old. Outer-loop link adaptation [1] closes the target gap, and faster reporting closes the age gap. What has been missing is the measured false-clear rate itself, which says how large the exposure is when a correction is absent, and when a correction has stopped being an optimization and become a requirement.

II. METHOD

Substrate. We use the 5G NR LDPC decoder in Sionna [2] as ground truth. For each MCS we measure a block-error curve $e_m(\gamma)$ by decoding 400 coded blocks per SNR point on a grid from -8 to 28 dB in 2 dB steps, over AWGN. The link then draws a per-TTI SNR trace from a 3GPP TR38.901 TDL-A channel at 3.5 GHz with 30 ns delay spread, and at each TTI the HARQ outcome of the selected MCS is drawn from e_m at the true SNR. This is a standard link-to-system mapping, built on measured decoder curves rather than a formula. The MCS set is the NR table (TS 38.214 Table 5.1.3.1-1), QPSK through 64-QAM at code rates of $1/5$ and up. Table I lists the parameters.

System model. Let $\gamma(t)$ be the true SNR in dB at TTI t and $\hat{g}(t)$ the estimate the transmitter holds. Fresh CSI reads $\hat{g}(t) = \gamma(t) + n(t)$ with $n \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 1$ dB. A report held with period P reads $\hat{g}(t) = \gamma(\lfloor t/P \rfloor)$ and ages between reports. Let $\rho_m(\beta) = \min\{\gamma : e_m(\gamma) \leq \beta\}$ be the SNR that MCS m needs to meet target β . The certificate selects, at a fixed reference target $\beta_0 = 10^{-1}$ with backoff Δ ,

$$m^*(\hat{g}) = \max\{m \in \mathcal{M} : \rho_m(\beta_0) + \Delta \leq \hat{g}\}, \quad \Delta=0 \text{ (naive)}. \quad (1)$$

The operating point is set by β_0 . The outcome is judged against the consumer's own target β , which may differ. The witness is the HARQ outcome, $\text{NACK}(t) \sim \text{Bernoulli}(e_{m^*}(\gamma(t)))$. A false-clear is a cleared MCS the channel cannot support at β , that is $\rho_{m^*}(\beta) > \gamma(t)$. We report its rate with two estimators. The refresh study (§IV) counts actual NACKs,

$$\widehat{\text{FC}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\text{NACK}(t)], \quad (2)$$

which is also what the outer loop consumes. The target study (§III) uses the expected rate $\text{FC}(\beta) = \frac{1}{T} \sum_t e_{m^*}(\gamma(t))$, because $T = 6000$ TTI cannot resolve a 10^{-3} tail by

TABLE I
LINK-LEVEL PARAMETERS (SEALED CELLS)

Parameter	Value
Carrier frequency	3.5 GHz
TTI duration	1 ms
Channel model	3GPP TR38.901 TDL-A, DS = 30 ns
Mean SNR	12 dB
Decoder / code	Sionna 5G NR LDPC (real decode)
MCS set	NR TS 38.214 Tbl 5.1.3.1-1 (QPSK-64QAM)
BLER curves	400 blocks/pt, SNR -8-28 dB @ 2 dB
CQI report noise σ	1 dB
Trace length	6000 TTI
eMBB / URLLC target	$10^{-1} / 10^{-3}$
URLLC-aware policy	conservative MCS + $K=3$ diversity
OLLA step $\delta_\uparrow$	0.1 dB ($\delta_\downarrow=0.9$ dB)
Main aging Doppler	200 Hz ($T_{\text{coh}} \approx 2.1$ ms, ≈ 62 km/h)
Refresh sweep f_D	10-400 Hz
Target-axis SNR sweep	3-21 dB

counting. Since (1) fixes the operating point at β_0 while the verdict uses β , one trace gives different answers to the eMBB consumer ($\beta=10^{-1}$) and the URLLC consumer ($\beta=10^{-3}$). That difference is what we measure.

URLLC-aware policy. It raises Δ and combines $K=3$ diversity branches. A backoff alone cannot reach 10^{-3} , because a single Rayleigh branch spends about 4% of its time in fades too deep for any MCS. The branches $\{\gamma_k(t)\}$ are independent TDL-A realizations from distinct seeds, standing in for frequency or spatial replicas rather than retransmissions, and a block fails only if all branches fail: $\text{FC}_{\text{aware}} = \frac{1}{T} \sum_t \prod_{k=1}^K e_{m^*}(\gamma_k(t))$. Independence makes this a best case. Correlated diversity fails together more often, so the true cost of URLLC safety is at least what we report.

OLLA. The correction of §IV does not trust $\hat{g}$. It selects $m^*(\hat{g}(t) - o(t))$ and moves the offset on HARQ feedback,

$$o(t+1) = o(t) + \delta_\uparrow \mathbf{1}[\text{ACK}(t)] - \delta_\downarrow \mathbf{1}[\text{NACK}(t)], \quad (3)$$

with $\delta_\uparrow = 0.1$ dB and $\delta_\downarrow = \delta_\uparrow(1 - \beta)/\beta$, so the loop settles where the NACK rate equals β . For held CSI the refresh floor is $P^*(f_D) = \max\{P : \widehat{\text{FC}} \leq \beta\}$, the longest report period that still meets the target.

Discipline. Each experiment is pre-registered. Numerical bars and manipulation checks are fixed before the graded run, the sealed record is content-hashed, grading uses seeds disjoint from those used during design, and failed or voided outcomes stay in the record [7]. The numbers below are the graded values (Table II).

III. THE RELIABILITY TARGET

A base station serving mixed traffic derives one CSI estimate per user and picks an MCS from it. Consider the same estimate read under two targets, 10^{-1} for eMBB and 10^{-3} for URLLC. We apply the naive CSI-derived MCS on a real NR link and measure the achieved BLER against each target. Then we compare a target-aware policy that sizes its protection to the stricter target.

Figure 2 shows the sealed result, three graded seeds of 6000 TTI each with fresh CSI. The naive certificate lands at a BLER

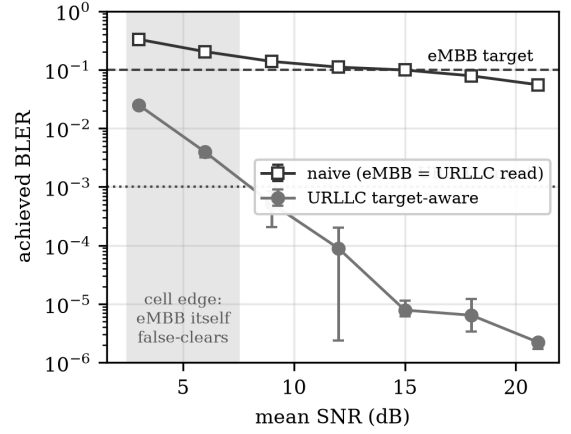


Fig. 1. The target gap across mean SNR (real NR, Sionna LDPC; sealed sweep, 3 seeds, whiskers min-max). The naive certificate rides the eMBB target, so the URLLC consumer sees roughly a hundredfold false-clear at every in-range SNR. Below about 6 dB (shaded) even the eMBB consumer is failed. The target-aware policy holds URLLC throughout.

near 0.112. The eMBB consumer reads this as sound, since it sits at the target the MCS was chosen for. The URLLC consumer reads the same 0.112 as a failure 112 times over budget. Each of those blocks breaks a reliability agreement, and nothing in the scheduler's view flags it, because the report looked fine. The target-aware policy uses the same CSI with a conservative MCS and the three independent branches of §II, and reaches 1.2×10^{-5} , within target. It pays in spectral efficiency and in diversity resources. So neither certificate serves both consumers well. The one sound for eMBB fails URLLC by 112 $\times$, and the one sound for URLLC overprovisions eMBB. The false-clear rate prices the difference.

The gap is not an artifact of the operating point we happened to pick. A separately sealed cell sweeps the mean SNR from 3 to 21 dB (Fig. 1). The certificate always selects the MCS that rides the eMBB target, so the URLLC consumer reads a false-clear of 0.08 to 0.14, between 77 and 142 times its budget, at every SNR in the working range of 9 to 18 dB, and still 54 times at 21 dB where the MCS table tops out. The target-aware policy holds URLLC across the whole range. Below about 6 dB the picture worsens in a different way. The lowest MCS is already too fast for the channel, so the certificate fails even the eMBB consumer it was tuned for, reaching a BLER of 0.33 at 3 dB. No margin repairs a certificate whose weakest format is already too aggressive.

IV. THE REFRESH HORIZON

The second axis is time. A report is fresh when issued and ages as the channel decorrelates. A certificate that was sound at report time false-clears once the channel has moved on. Here we count actual HARQ NACKs, Eq. (2). At 200 Hz Doppler, about 62 km/h at 3.5 GHz, with a report every 20 TTI, the held estimate is off by 5.7 dB on average. Aging dominates that error; report noise contributes about 1 dB. The naive certificate then runs at a BLER near 0.36 against a 0.10 target, a false-

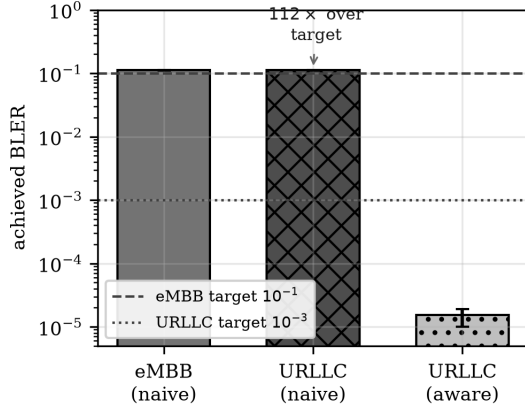


Fig. 2. One report, two verdicts (real NR, Sionna LDPC; 3 sealed seeds, 6000 TTI each; bars are means, whiskers min-max). The same CSI-derived MCS meets the eMBB target and misses the URLLC target by 112 $\times$. The target-aware policy meets URLLC.

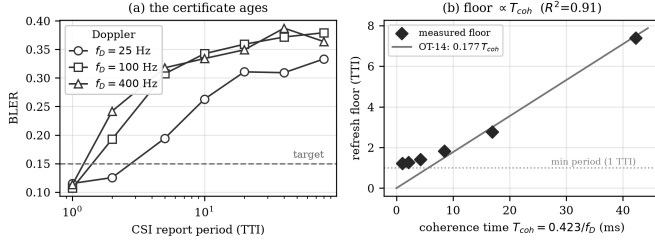


Fig. 3. The refresh horizon (real NR, Sionna LDPC; sealed sweep). (a) BLER grows with report period, faster at higher Doppler. (b) The refresh floor is proportional to the Clarke coherence time ($0.177 T_{\text{coh}}$, $R^2 = 0.915$) and saturates at one TTI at high Doppler.

clear on roughly a quarter of transmissions. OLLA, Eq. (3), corrects the same held report with HARQ feedback and holds 0.10. Figure 3(a) shows the aging across mobility.

The practical question is how often to report. We sweep Doppler from 10 to 400 Hz, which spans coherence times from 42 ms down to 1 ms, and report periods from 1 to 80 TTI. The refresh floor $P^*(f_D)$ is the longest period that keeps BLER at target. A least-squares fit through the origin against the Clarke coherence time $T_{\text{coh}} = 0.423/f_D$ gives

$$P^* \approx 0.177 T_{\text{coh}}, \quad R^2 = 0.915 \quad (4)$$

(Fig. 3(b)). The low-mobility points dominate the fit. At high Doppler the floor stops at one TTI, the shortest period the frame allows, and that stop matters. Above roughly 200 Hz no reporting cadence keeps the naive certificate at target, so a HARQ-driven correction is no longer an optimization but a requirement. Below that threshold the fit is a usable rule. Report again before the channel decorrelates by about $0.18 T_{\text{coh}}$, rather than on a fixed schedule.

V. ONE BUDGET, TWO FLEETS, OPPOSITE VERDICTS

The two axes above share a stronger property: the verdict can invert. Consider two user fleets whose failures come from

TABLE II
SEALED PER-SEED RESULTS (ACHIEVED BLER)

(a) Reliability target (§III), fresh CSI				
seed	eMBB	URLLC-naive	URLLC-aware	
20260824	0.1119	0.1119	1.7×10^{-5}	
20260825	0.1103	0.1103	1.9×10^{-5}	
20260826	0.1125	0.1125	$< 10^{-5}$	
(b) Refresh horizon (§IV), $f_D=200$ Hz, 20-TTI				
seed	naive	OLLA	fresh	CQI err
20260823	0.366	0.101	0.116	5.61 dB
20260824	0.340	0.102	0.115	5.86 dB
20260825	0.373	0.101	0.113	5.76 dB
Seed IDs are campaign date-stamps. (a) expected rate; (b) empirical NACK rate.				

Seed IDs are campaign date-stamps. (a) expected rate; (b) empirical NACK rate.

different parts of the radio state. The M-fleet runs at 200 Hz Doppler and 12 dB mean SNR, so its blocks die of CSI aging. The S-fleet runs at 10 Hz and 7 dB, so its blocks die in deep fades. Now take two margin policies with the same fleet-mean budget of 1.5 dB: policy A allocates by Doppler, policy B allocates the same total by SNR deficit. The M-fleet does better under A (NACK rate 0.22 against 0.36), and the S-fleet does better under B (0.16 against 0.27), on every seed. The fleet-mean NACK rate sits within 0.02 for the two policies, so the aggregate cannot order the pair for anyone. A control run sharpens the mechanism: reading one fleet at two reliability thresholds (a shifted required-SNR table) under redistributed margin produces no inversion at all, because a threshold shift leaves both consumers reading the same projection of the state. That is also why §III showed dominance rather than inversion: eMBB and URLLC are a threshold pair. The inversion needs consumers that read different axes, mobility against depth, and a mixed cell has both. A scheduler that tunes its reporting and margin to one fleet-level KPI is optimizing a number that no user experiences.

VI. RELATED WORK

Link adaptation, CSI aging and outer-loop correction are old subjects. OLLA [1] holds a BLER target under imperfect or stale CSI by adjusting an SINR offset from HARQ feedback. It is exactly the correction our refresh result quantifies. Short-packet analysis [3] establishes the reliability regime where the eMBB-to-URLLC gap lives. Deployed systems configure CQI reporting at a fixed periodicity [4]; our refresh floor turns that period into a coherence-time budget instead. Learned CSI feedback trained on reconstruction error [6] is an active direction. In our own controlled tests the objective gap it might suggest did not appear at practical operating points, so we claim nothing about it. We do not offer a better estimator, scheduler or offset rule. We offer the measurement, and the two dependences and one law it exposes.

VII. DISCUSSION AND CONCLUSION

The unit of correctness in link adaptation is not the CSI report. It is the pair of report and consumer. Margin and reporting cadence sized to the consumer's target and coherence time keep a certificate honest. Sized to a nominal channel they do not, and the cost is measurable: a factor of 112

for a mismatched target, and a coherence-scaled horizon past which a stale report cannot be saved by any cadence. The evidence stands on Sionna link-level simulation with real LDPC decoding. Over-the-air HARQ traces are the natural next step.

The evidence has limits worth stating. It uses one TDL-A profile at 30 ns delay spread, one MCS table topping out at 64-QAM, and one mobility for the target axis. The refresh law is fit across 10 to 400 Hz, and the target gap is measured across 3 to 21 dB, including the cell edge where the certificate fails its own eMBB consumer. The factors we left out push the numbers the same way. 256-QAM adds steeper cliffs that worsen aging, and correlated diversity raises the aware policy’s failure rate. On implementation, the refresh floor needs no new machinery. Event-triggered CSI reporting with a relative threshold already exists in NR [4] and is an instance of send-on-delta sampling [5]. In exploratory runs such a trigger reproduced the floor constant, near $0.2 T_{\text{coh}}$, but our sealed confirmation attempt was voided by a control-sanity check at very low Doppler, so we report that outcome and claim nothing beyond the prior art. The method itself is not specific to radio. The same consumer-relative failure shows up in optical quality-of-transmission and in database replication [7]. The results here stand on the NR CSI certificate alone.

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